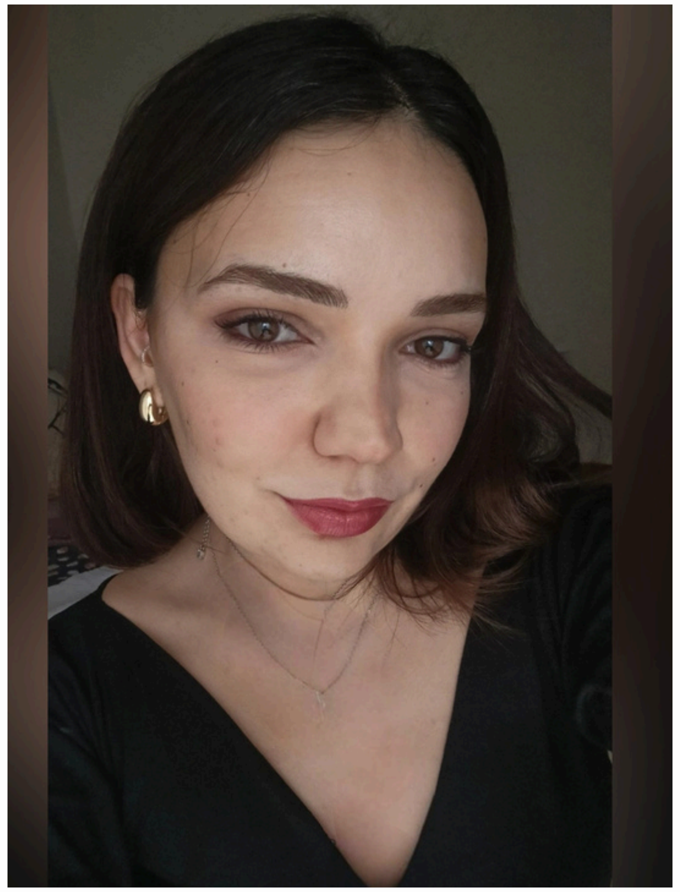

AYŞE KADRİYE YAZICI | Embedded Software Engineer

EDUCATION

Electrical-Electronics Engineering, Bachelor's Degree | Ondokuz Mayıs University• 2019 – 2023
GPA: 3.00 / 4.00

SKILLS

Embedded Systems	ARM Cortex-M, Bare Metal, FreeRTOS, Embedded C
Communication & Control	UART, SPI, I2C, CANBUS, MODBUS, MQTT, RF
Platforms & Tools	STM32F4, F1 - ESP32 - TI MCU - PeakCAN - cubeIDE ESP-IDF - Visual Studio
Test & Validation	Oscilloscope, Lojik Analizör, JTAG/SWD(ST-Link, J-Link)
Version Control	Git-Github, Fork, TortoiseGit

CAREER GOALS

I aim to bring my experience in embedded software development to safety-critical systems in the defense, aerospace, and automotive sectors, taking an active role in projects that require real-time control, multi-ECU communication architectures, and high reliability. I aspire to become an engineer specialized in processes ranging from system integration to verification, with a strong command of industry standards such as DO-178C/MIL-STD/ISO 26262. In the long term, I want to shape my career as an embedded systems engineer who contributes to the development of domestic and national defense/aerospace technologies and who can take on team leadership responsibilities.

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WORK EXPERIENCE

Embedded Software Engineer | Remergen Mekatronik • Bilişim Vadisi • November 2024 - April 2026

- *Developed real-time embedded software based on STM32 for mechatronic systems in the automotive, industrial, and white goods sectors.*
- *Designed low-level drivers and multi-ECU CAN Bus communication architecture for distributed control systems using SPI, I2C, UART, CAN, NVIC, DMA, ADC, TIMER, and PWM peripherals.*
- *Developed actuator control algorithms including PWM-based and closed-loop control strategies to ensure deterministic system behavior.*
- *Performed system integration, debugging, and validation using oscilloscope and logic analyzer; contributed to DFMEA/PFMEA reliability studies.*

Embedded Software Developer (Intern) | BAYKAR — Autopilot Development Systems Team • August 2023 – January 2024

- *Took part in embedded software development projects running on ARM-based microprocessors within the Autopilot Development Systems unit.*
- *Gained firsthand experience with aviation-grade embedded systems development processes, team working discipline, and quality standards.*

CRANE CONTROL SYSTEMS OVERVIEW

- *I took part in implementing and testing the communication infrastructure and embedded software of a multi-ECU CAN Bus architecture — consisting of six separate electronic control units — developed at Remergen Mekatronik for basket platform cranes, telescopic/articulated boom platforms, and crane-mounted systems.*

CENTRAL CONTROL UNIT (CRANE MAIN CONTROLLER)

- *Developed the firmware for the control unit that serves as the system's main brain.*
- *Implemented PWM and ON/OFF valve driver outputs and 4-20mA/0-5V/0-32V analog-digital input management for 56/121-pin variants.*
- *Developed a dual-channel 50kb–1Mbit CANBus communication architecture and a configuration/operation data logging mechanism on the internal 64Mb EEPROM.*
- *Designed the communication protocol by defining unique CAN message IDs (arbitration IDs) for each ECU, following prioritization and collision-avoidance logic.*
- *Developed data packaging functions that encode/decode sensor data, commands, status information, and fault codes into the standard CAN data frame format.*
- *Ensured detection and proper error handling of corrupted or lost data packets through checksum/CRC verification and timeout controls.*

TILT ANGLE & DISTANCE SENSOR (CRANE TILT ANGLE & DISTANCE CONTROLLER)

- *Developed the firmware for the sensor unit that measures the boom's working angle and boom extension length on a single unit and transmits this data to the central controller via CANBUS.*
- *Implemented distance measurement algorithms for the product line covering measurement ranges from 4m to 30m.*
- *Achieved data stabilization through 2-axis angle (-10°/+110°) measurement, a mapping algorithm that scales the data to the appropriate range, and median and low-pass filter functions.*
- *Implemented application-specific reprogrammable output logic and data transmission via CANBUS with the central control unit and HMI interface.*

REMOTE CONTROL UNIT (CRANE REMOTE CONTROLLER)

- *Developed the software for the remote control unit that unifies different operator input types — 2-way, 4-way, joystick, finger joystick, and thumbwheel joystick — on a single platform.*
- *Read analog signals from the joystick axes via ADC and normalized them into usable control values through a mapping function that compensates for dead-zone and non-linear characteristics.*
- *Transmitted the processed data to the main processor in real time via CANBus or optional 2.4GHz RF; the main processor converted this data into a PWM signal to drive the valve outputs for proportional movement of the hydraulic actuators.*

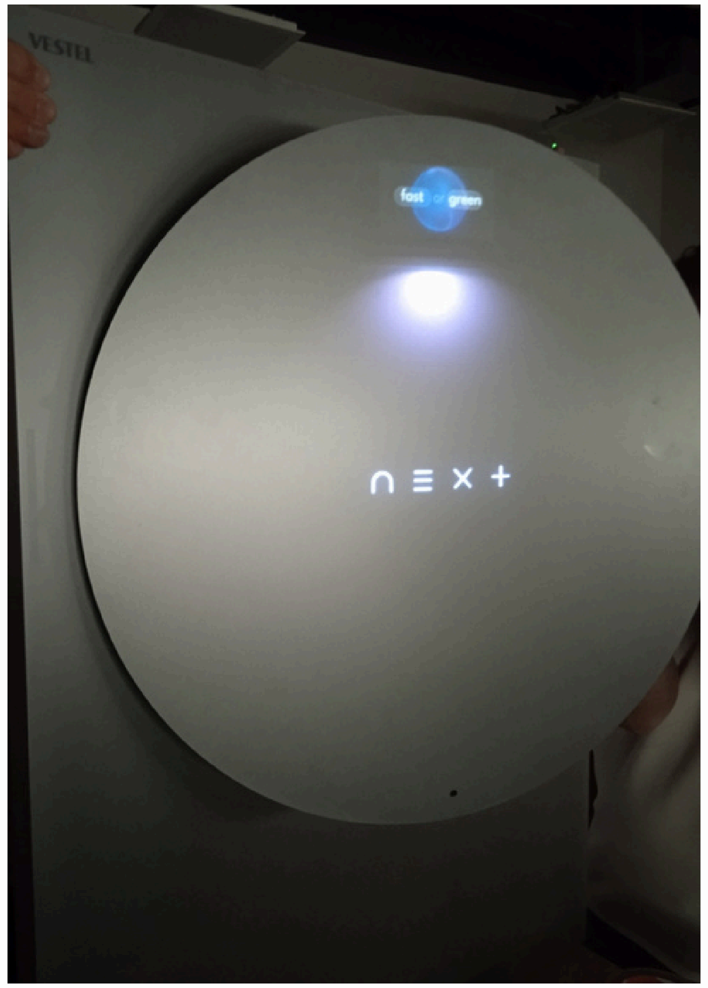
BASKET BALANCE CONTROL UNIT (CRANE BASKET BALANCE CONTROLLER)

- *Developed the firmware for the ECU that detects the basket angle via an internal angle sensor and provides automatic balancing.*
- *Implemented a control algorithm that ensures real-time, automatic balancing of the basket based on angle data, using PWM valve and PWM coil driver outputs.*
- *Developed an auxiliary controller architecture that shares sensor data with and executes commands from the central controller via CANBus; ensured independent reprogrammability from the rest of the system using an internal EEPROM.*

Vestel NEXT Washing Machine — Trade Show Product 🏆 iF Design Award 2026

Took part in the embedded software and user experience components of the NEXT AI Series washing machine, developed in collaboration with Vestel and awarded the iF Design Award 2026.

- Developed the software controlling the RGB LEDs that create the device's ambient lighting, dynamically adjusting them based on the active function (e.g., different colors/effects for 'fast' or 'green' mode).
- Developed the user-interface software by programming the device's Nextion touchscreen, enabling selection and execution of machine functions.
- Designed the communication and function-status synchronization between the hardware (RGB LED driver + Nextion screen) and the main control logic.



COMPETITIONS - COURSES

- TEKNOFEST Agricultural Unmanned Ground Vehicles Competition — Turkey 2nd Place • 2023
- TEKNOFEST Unmanned Aerial Vehicles Competition — Finalist • 2023
- TEKNOFEST Unmanned Aerial Vehicles Competition — Finalist • 2022
- ISO 9001:2015 Quality Management System Basic Training — SigmaCert Global
- Erhan KONAK - Microcontroller Driver Development Course - Udemy
- FastBit Embedded Brain Academy - ARM Cortex M Microcontroller DMA Programming Demystified - Udemy
- FastBit Embedded Brain Academy - STM32Fx Microcontroller Custom Bootloader Development - Udemy

CONTACT

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Sincerely,
Ayşe Kadriye YAZICI